Kasin Ito-Sato

Education: Loyola Marymount University

Year: Junior

Hometown: Pearl City, HI

Role: Mechanical Engineering Intern

Supervisor: Aaron Ginoza

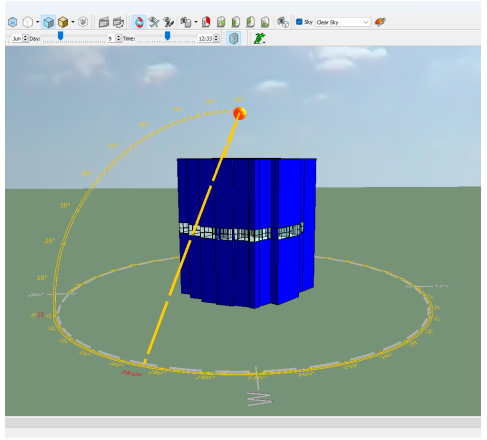


GOALS

- Improve my resume
- Visit a job site
- Perform a cooling load calculation
- Gain understanding of residential MEP systems
- Confirm interest in building systems consulting industry

HAWAII PRINCE WAIKIKI SPA RENOVATION

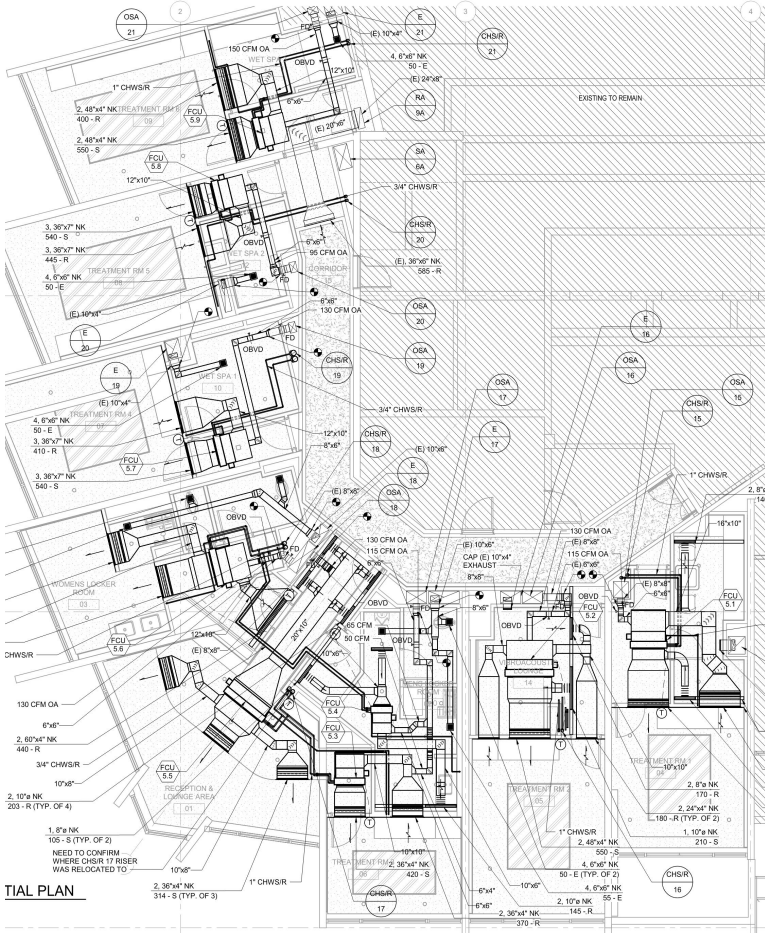
- Ductwork
- IES
- Room data sheet



The Design Engineer should complete the ventilation air volume with the engineer of record

							VENTILATION AIR VOLUME				DEMAND CONTROLLED VENTILATION	
Room	Room Name	Area	People	Temp.	RH	Room Type	Room Type	Vent	Vent	#/1000 ft²	Served By System With:	Exceptio
1	RECEPTION & LOUNGE AREA	407	10	75	60	Office >300ft^2	Lobbies	5	0.06	150	None	None
2	MEN'S LOCKER ROOM	192	-	75	60	Office (>150 and <300 ft^2)	Multipurpose	5	0.06	120	None	None
3	WOMEN'S LOCKER ROOM	283	-	75	60	Office (>150 and <300 ft^2)	Multipurpose	5	0.06	120	None	None
4	TREATMENT ROOM 1	379	2	75	60	Office (>150 and <300 ft^2)	Health	20	0.06	40	None	None
5	TREATNENT ROOM 2	370	2	75	60	Office (>150 and <300 ft^2)	Health	20	0.06	40	None	None
6	TREATMENT ROOM 3	127	2	75	60	Office (<150 ft^2)	Health	20	0.06	40	None	None
7	TREATMENT ROOM 4	255	2	75	60	Office (>150 and <300 ft^2)	Health	20	0.06	40	None	None
8	TREATMENT ROOM 5	242	4	75	60	Office (<150 ft^2)	Health	20	0.06	40	None	None
9	TREATMENT ROOM 6	280	4	75	60	Office (>150 and <300 ft^2)	Health	20	0.06	40	None	None
10	DRY SAUNA	94				Convention Ceneter -- Exhibit Space	N/A	0	0	0	None	None
11	ONSEN ROOM	84				Convention Ceneter -- Exhibit Space	N/A	0	0	0	None	None
12	STEAM ROOM	106				Convention Ceneter -- Exhibit Space	N/A	0	0	0	None	None
13	OFFICE	107	4	75	60	Office (<150 ft^2)	Office space	5	0.06	5	None	None
14	LOUNGE	142	2	75	60	Lounge/Breakroom (All other lounges/	Lobbies	5	0.06	150	None	None
	VESTIBULE	66		75	60		N/A	0	0	0	None	None
	Total	421	6									

NOTES:



ROOM DATA SHEET ADDITIONS

- Learning
- Fundamentals
- Team help
- Efficiency

Demand Controlled Ventilation (DCV) Excel Sheet Walk-through

1. DCV background
2. How the DCV excel sheet works
3. Examples

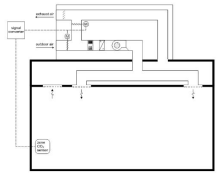
You must refer to ASHRAE 90.1 2016 - Section 6.4.3.8 for criteria and exceptions.

Criteria:

1. Required for spaces >500 ft^2
AND

2. >=25 people per 1000 ft^2 of floor area
AND

3. Served by ONE OR MORE of:
   a. An air side economizer
   b. Automatic modulating OA damper
   c. Design outside airflow >3000 cfm



Exceptions:

1. Systems with EA energy recovery complying with Section 6.5.6.1.
2. Multiple-zone systems w/o DDC of individual zones communicating with a central control panel.
3. Systems with a design outdoor airflow less than 750 cfm.
4. Spaces where >75% of the space design outdoor airflow is required for makeup air that is exhausted from other spaces.
5. Spaces with one of the following occupancy categories as defined in ASHRAE Standard 62.1: correctional cells, daycare sickroom, science labs, barbers. beauty and nail salons, and bowling alley seating.

1

Building Pressurization Excel Sheet Walk-through

1. Background on building pressurization
2. How the excel sheets works
3. Examples

Low Pressurization
- excess infiltration
- unfiltered air
- humid air condenses causing mold

High Pressurization
- doors inconveniently propped
- large waste in energy
- conditioning too much OA

<= 0.02 in w.c. = under pressurized
0.05 in w.c. = ideal
>= 0.1 in w.c. = over pressurized

Building Envelope Construction
- never perfect
- falls within building tightness: tight, average, loose, very loose
  - # of windows and door
  - contractors building history
  - engineering judgment
  - area of flow to are of exterior space

Building Leakage Rate
- amt of air that escapes

Do not rely on tight construction, can always design to reduce OA if over-pressurized

Why not use "10% rule of thumb?"

Ex: 400 ft X 100 ft open office w/ 15 ft walls and a default occupant density of five people per 1000 ft^2 = 200 people.

Vbz = 5 cfm/per x 200 people + 0.06 cfm/ft^2 x 40000 ft^2 = 3400 cfm

Vexcess = 3400 cfm x 10% = 340 cfm

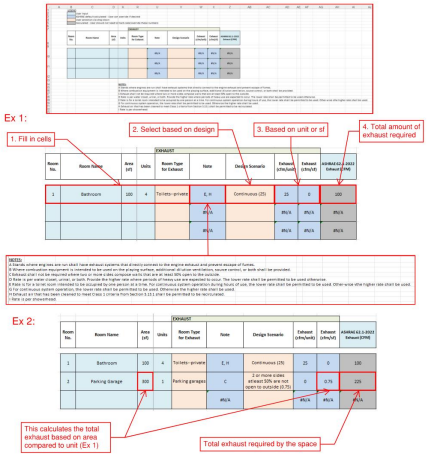
delPavg = (340 cfm / (2610 x 2.55 ft^2))^2 = 0.0026 in. w.c.

0.0026 in w.c. means the building is **VERY** under-pressurized (0.0026 < 0.02).

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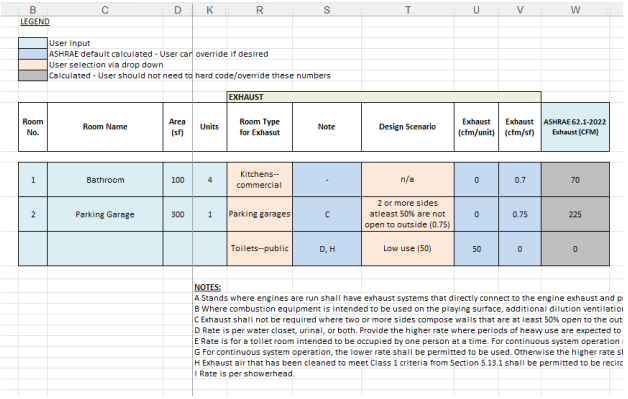
Exhaust Rate Excel Sheet Walk-through

1. How the exhaust rate excel sheet works
2. Examples



LEGEND
- User Input
- ASHRAE default calculated - User can override if desired
- User selection via drop down
- Calculated - User should not need to hard code/override these numbers

Room No.	Room Name	Area (sf)	Height (ft)	People	VENTILATION AIR VOLUME: Room Type for Ventilation	Vent (cfm/per.)	Vent (cfm/sf)	#/1000 ft² (#/100 m²)	Served By System With:	Exceptions (NO DCV)	DCV?	Air Distribution Configuration	E_z Value	Activity (MET Level)	MET	Room CO_2 Concentration	ASHRAE 62.1-2007 ventilation (CFM)	OA w/ E_z Design	OA Low Setting
1	CONFERENCE ROOM	2,000	10	100	Conference/meeting	5	0.06	50	Air economizer	None	Yes	Ceiling supply of warm air 15°F or more above space temp and ceiling return	0.8	Reading and writing, seated	1	1,483.87	620	775	150
2	OPEN OFFICE	1,500	10	15	Office space	5	0.06	5	Any combo of above	None	No	Ceiling supply of cool air	1	Typing	1.1	1,240.00	165	165	No DCV
3	SCHOOL AUDITORIUM	7,000	40	640	Auditorium seating area	5	0.06	150	Any combo of above	None	Yes	Floor supply of warm air and ceiling return	0.7	Seated, quiet	1	1,439.56	3,620	5,171	600



SITE VISIT



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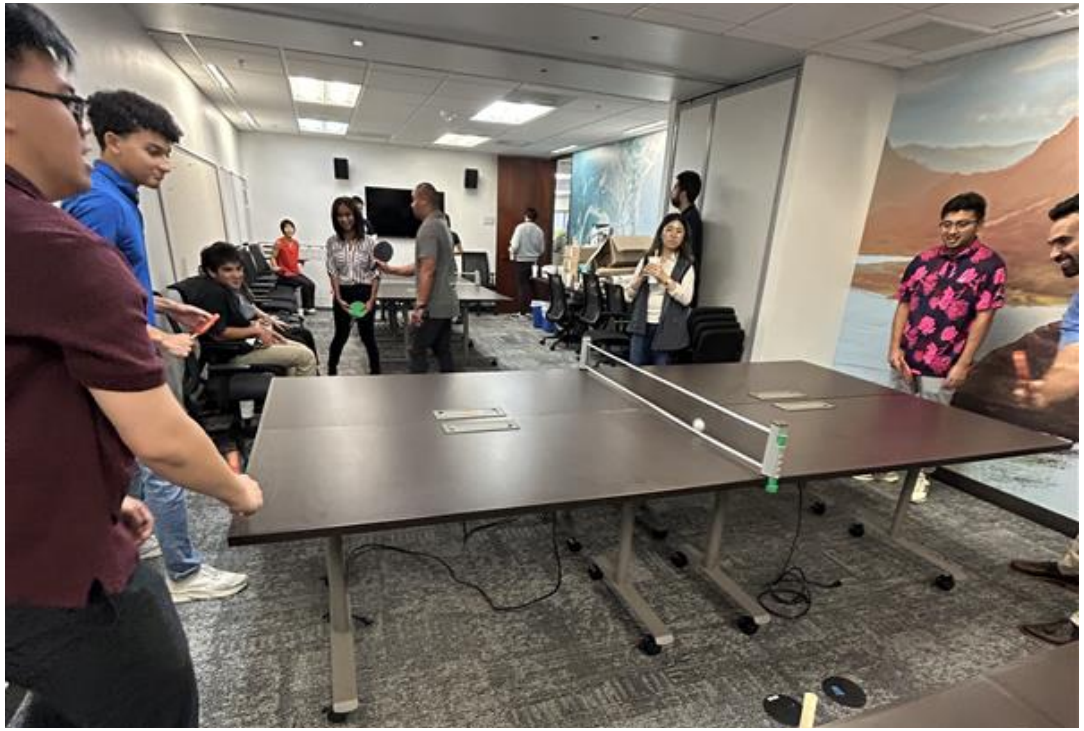


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Ulana

FAVORITE MOMENTS



DPN Events



ASHRAE X IECC

WORDS OF APPRECIATION

Aaron Ginoza

Kurt Metrose

The whole WSP office and MEP team!